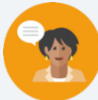


10.2 Arc Length

Lesson Introduction

 Benchmark	MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.		 Learning Targets	- I can define arc length. - I can explain how the length of the arc is proportional to the ratio of the central angle to the whole circle, measured in degrees.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is an arc length of a circle? - How is the length of an arc proportional to the ratio of the central angle to the whole circle? - What is the relationship between the length of an arc and the circumference of a circle?			
 Lesson Narrative	In this lesson, students will learn that the length of an arc varies directly with the central angle, and is a fraction of the circumference of the entire circle. Students will discover that the ratio of the arc length to the circumference of the circle is equal to the ratio of the central angle to the whole circle.			
 Component Summary	10.2.1 Warm-Up (~5 min.)	Examining an image of the Chandelier Tree (<i>SE p. 125</i>)	10.2.3 Activity (10 min.)	Matching central angles and ratios of arc length to circumference (<i>SE p. 127</i>)
	10.2.2 Exploration (15-20 min.)	Discovering the relationship between arc length and central angles (<i>SE p. 125</i>)	10.2.4 Exploration (10 min.)	Discovering the relationship of central angles between circles of different radius measures (<i>SE p. 128</i>)
	10.2.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p.129</i>)		
 Resources	Glossary		Materials	
	Key Terms: N/A Additional Glossary: Arc Length		- Cards for Card Sort (optional) String	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may have difficulty simplifying fractions to see the connection between the ratio of the arc length to the circumference, and the ratio of the central angle to the whole circle.	- Facilitate a conversation to distinguish between arc measure and arc length. - Students should not be using the formal formula for calculating the arc length yet. - If time is an issue, then consider assigning 10.2.4 Exploration for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect The lesson provides students with the opportunity to discover the relationship between the ratio of the central angle to the whole circle, and the ratio of the arc length to the circumference. Encourage students to compare and connect their findings to the discoveries from the previous lesson about sector area to the new discoveries made in this lesson. Allow students the chance to review examples from the lesson, so they can deepen their understanding about central angles.	MA.K12.MTR.4.1: Discussion and Reflection Use the 10.2.2 Exploration and the 10.2.3 Activity to encourage students to share their mathematical thinking with peers. Provide encouragement by creating a safe environment for students to communicate their mathematical ideas and accept feedback. Encourage students to use mathematical vocabulary during discussions.
 Differentiate Instruction	This lesson provides multiple entry points for visual, auditory, and kinesthetic learners. Students draw their own conclusions and discuss their rationale with peers and teachers, providing an opportunity for auditory learners to verbalize their conception. Consider providing string for kinesthetic learners to use for a hands-on interaction for learning about arc length. Images are shown throughout the lessons for visual learners to see the connection between the central angle and the arc length.	
Lesson Notes:		

10.2.1 Warm-Up: Notice and Wonder

Component Overview: Students will make observations about what they notice and wonder about the image. Students will estimate the diameter of the base of the tree and justify their answer.



10.2.1 Warm-Up: Notice and Wonder

The Chandelier Tree, located in Leggett California, is part of a drive-thru tree park. The tree is believed to have been carved in the early 1930's by Charlie Underwood.

1. What do you notice?

Answers will vary. I noticed that the trunk of the tree is very wide. I noticed that tree looks very sturdy. I notice that the tree is gigantic.



2. What do you wonder?

Answers will vary. I wonder how old the tree is. I wonder how they knew the tree could still stand after that much being carved out. I wonder how that hole got there.

3. Estimate the diameter of the tree at the base.

Answers will vary. I estimate the diameter of the tree to be 12 feet. I estimate the diameter of the tree to be 20 feet.

4. Explain why you chose that number as your estimate.

Answers will vary. I chose this number because it needs to be large enough for a car to go through. I chose this number because I estimated the length of the car in the photo (roughly 15 feet) and added a few more feet since the tree diameter is larger than the car.



Notice and Wonder

Utilize the built-in "Notice and Wonder" routine while students analyze the information and the image. Position students to share their findings with a partner to expand their curiosity.

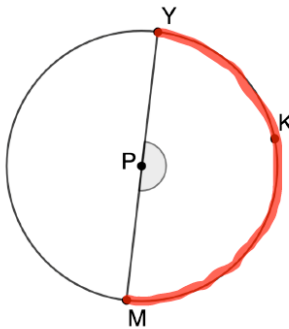
10.2.2 Exploration: Arc Length and Proportional Relationships

Component Overview: In this Exploration, students will build a connection between the central angle and the fraction of the circumference of a circle. Students will discover that the arc length is the distance along the circumference of a circle, and that the measure of the arc is proportional to the ratio of the central angle to the whole circle.



10.2.2 Exploration: Arc Length and Proportional Relationships

1. Circle P with point K on the circle and diameter $\overline{YM}$ is shown.



- a. What is $m\angle YPM$?
 180°
- b. Determine the ratio of the measure of the central angle, $\angle YPM$, to the whole circle. Simplify the ratio.
 $\frac{180^\circ}{360^\circ} = \frac{1}{2}$
- c. An arc of a circle is the portion of the circumference between points. Point K is a point on Circle P . Discuss with your partner which part of the circle is arc YKM . Then, trace over this portion of the circle.
Students do not need to write an explanation here, but a sample response is provided for reference.
Since arc YKM of the circle is 180° , arc YKM is half of the circle.
Tracing is on the image of Circle P above.
- d. What fraction of the circle's circumference does arc YKM represent?
 $\frac{1}{2}$



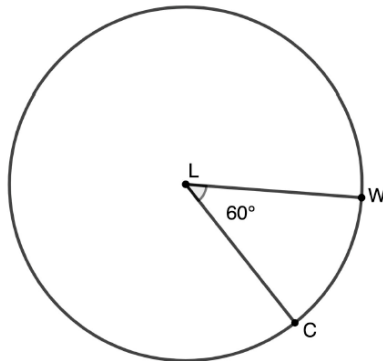
Consider prompting questions to help students make sense of the problems, as a check for understanding.

- What is a central angle of a circle?
- What is the circumference of a circle?

10.2.2 Exploration: Arc Length and Proportional Relationships (continued)

Arc YKM can also be notated as $\widehat{YKM}$.

2. Circle L is shown where points W and C are on the circle.



- a. What is the ratio of the central angle, $\angle CLW$, to the whole circle? Simplify the ratio.
 $\frac{60^\circ}{360^\circ} = \frac{1}{6}$
- b. What fraction of the circle's circumference does $\widehat{WC}$ represent?
 $\frac{1}{6}$
3. Verify your answers to problems 1 and 2 with your partner and resolve any differences, if necessary.
Answers will vary.
 $\frac{180}{360} = \frac{1}{2}$, so, any arc that is 180° , will be half the circumference of a circle. Any arc that is 60° , will be $\frac{1}{6}$ of the total circumference of a circle.



Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the relationship between the ratio of the central angle to the whole circle, and the fraction of the circle's circumference the arc makes. This will allow students the chance to make observations to deepen their understanding of arc lengths before discussing with a partner.



Students do not need to write an explanation here, but a sample response is provided for reference.

10.2.2 Exploration: Arc Length and Proportional Relationships (continued)

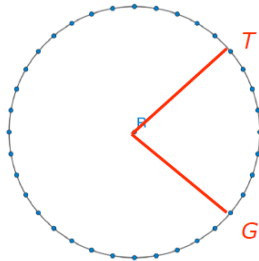
4. With your partner, discuss any relationships you observe. Summarize your discussion.

Answers may vary.

The ratio of the central angle to the whole circle is equivalent to the fraction of the circumference that the arc represents.

A fraction of a circle's circumference that an arc represents is the same as the ratio of the central angle to the whole circle.

5. On Circle R , the distance between each point measured along the circumference is 1 inch.



- a. Mark a point anywhere along the circle's circumference as point T . Then, starting at point T , mark an 8-inch section along the circle's circumference. Label the other endpoint of the 8-inch section as point G .

Answer on image above. Point locations will vary.

- b. What fraction of the circle's circumference does the length of $\widehat{TG}$ represent?

$$\frac{8\text{in}}{36\text{in}} = \frac{2}{9}$$

- c. Draw the central angle $\angle TRG$.

Answer on image above. Locations may vary.

- d. What is the measure of the central angle $\angle TRG$?

$$\frac{2}{9}(360^\circ) = 80^\circ$$



Consider giving students string to place on the circumference of the circle to create arc TG . This hands-on approach will help kinesthetic and visual learners visualize the arc in relation to the circumference of the circle.



Clarify, Critique, Connect

Consider providing different lengths of string to engage partners in a discussion about their findings. Students can trade their responses with their partner, then ask their partner clarifying questions, if needed. This process will allow students to think more analytically about the relationship between the arc length and the central angle.

10.2.2 Exploration: Arc Length and Proportional Relationships (continued)

- e. What is the ratio of the central angle to the whole circle?

$$\frac{80^\circ}{360^\circ} = \frac{8}{36} = \frac{2}{9}$$

- f. Discuss with your partner the relationship between the length of the arc and the ratio of the central angle to the whole circle.

Students do not need to write an explanation here, but a sample response is provided for reference.

The ratio of the length of the arc to the circumference of the circle is the same as the ratio of the central angle to the whole circle.

6. Complete the statements.

The ☐ arc length is the distance along the part of

- ☐ diameter
☐ radius

a circle's circumference or along any curve. The ratio

of the length of an arc to the circumference of a circle

is ☐ the inverse of the ratio of the central angle to

- ☐ proportional to

the whole circle.



Consider asking the following to check for understanding before moving on:

- How is an arc length related to the circumference of a circle?
- What is the relationship between the measure of an arc and the ratio of the central angle to the whole circle?

10.2.3 Activity: Matching Central Angles and Ratios of Arc Length to Circumference

Component Overview: Students will work with their partner to match each circle to the correct ratio of the arc length to the circumference of the circle indicated on the card.



10.2.3 Activity: Matching Central Angles and Ratios of Arc Length to Circumference

1. Your teacher will give you two sets of cards for this activity. One set shows circles with a central angle. The other set of cards is a set of ratios of the arc length to the circumference.
 - a. Work with your partner to match each circle card to the corresponding ratio indicated on the lettered card. Record the matches in the table.

Card Letter	Ratio of the Arc Length to the Circumference	Card Letter	Ratio of the Arc Length to the Circumference
A	$\frac{1}{6}$	F	$\frac{5}{18}$
B	$\frac{1}{3}$	G	$\frac{2}{9}$
C	$\frac{7}{18}$	H	$\frac{1}{2}$
D	$\frac{1}{8}$	I	$\frac{5}{12}$
E	$\frac{1}{12}$	J	$\frac{5}{24}$

- b. Summarize how you and your partner were able to match the circles to a corresponding ratio without being given the circumference. *Answers may vary. The circles were matched to the corresponding ratios by finding the ratio of the central angle to the whole circle, which is equivalent to the ratio of the arc length to the circumference.*

- c. Complete the statement.
The ratio of the central angle to the whole circle is the

☐ inverse of the ratio of the length of an arc to
☐ same as the circumference.



Consider asking the following while monitoring students during the activity to check for understanding:

- How is the central angle used to find the ratio of the arc length to the circumference?
- Will you always need the central angle to find the measure of an arc of a circle? Explain.



Consider challenging students who completed their work early to create a formula that can be used to find the measure of an arc of a circle. Encourage students to include the central angle in the formula.

10.2.4 Exploration: Arc Lengths With the Same Central Angle and Different Radii

Component Overview: Students will discover that when circles have the same central angle, the ratio of the arc length to the circumference will equal the ratio of the central angle to the whole circle. Students will justify their discovery by calculating the ratios for arc lengths to the circumference, and the ratio of the central angle to the whole circle.



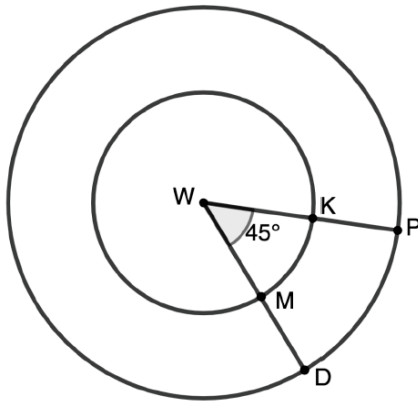
10.2.4 Exploration: Arc Lengths with the Same Central Angle and Different Radii

- Points D and P lie on Circle W . The radius $\overline{WP}$ measures 5 inches (in.).

A smaller Circle W with a radius measure of 2 in. rests inside the larger Circle W , where points M and K lie on the smaller circle.

The smaller Circle W has central angle $\angle KWM$ that measures 45° and the larger Circle W has central angle $\angle PWD$ that measures 45° .

Use the information to complete the following.



- $\widehat{MK}$ and $\widehat{DP}$ are different lengths. Discuss with your partner whether you think the ratio of the arc length and circumference will be the same as or different than the ratio of the central angle of the whole circle.

Students do not need to write an explanation here, but a sample response is provided for reference. We think the ratio will be the same because they have the same central angle.



Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the image of smaller Circle W inside larger Circle W . This will provide students with the opportunity to analyze the image before discussing with a partner.

10.2.4 Exploration: Arc Lengths With the Same Central Angle and Different Radii (continued)

- b. Andrew and Sarah are discussing if the ratio of the arc length to the circumference is the same as the ratio of the central angle to the whole circle for both the smaller and larger circles.

Sarah says, "Yes. They will have the same ratios because they have the same central angle."

Andrew says, "No. They do not have the same ratios because they are different circles with different radii."

Explain who you agree with.

Answers will vary. Sarah is correct, the ratio of the central angle to the whole circle is the same as the ratio of the arc to the circumference.

- c. Complete the table to support your reasoning in Part B.

	Ratio of Arc Length to Circumference	Ratio of Central Angle to Whole Circle
Circle W with radius 2 in.	$\frac{1}{8}$	$\frac{45^\circ}{360^\circ} = \frac{1}{8}$
Circle W with radius 5 in.	$\frac{1}{8}$	$\frac{45^\circ}{360^\circ} = \frac{1}{8}$

- d. Explain whether your findings in Part C support your answers to Parts A and B. If not, explain why your answers to Parts A and B are incorrect.

Answers will vary. My findings support my answers to Parts A and B, this is because the ratios are the same.

My findings do not support my answers to Parts A and B. This is because the ratio of the arc length to the circumference were not the same as the ratio of the central angle to the whole circle.



To strengthen the conception of concentric circles, consider asking students:

- Will all circles with the same central angle have the ratio for arc length to circumference equal the ratio of the central angle to whole circle?

10.2.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



10.2.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.



Consider collecting data from the cool down to gain insight on student understanding of the lesson. If many students indicate misconceptions about arc length, a review may be necessary prior to Lesson 3.